Response time :-

Turnaround time :-

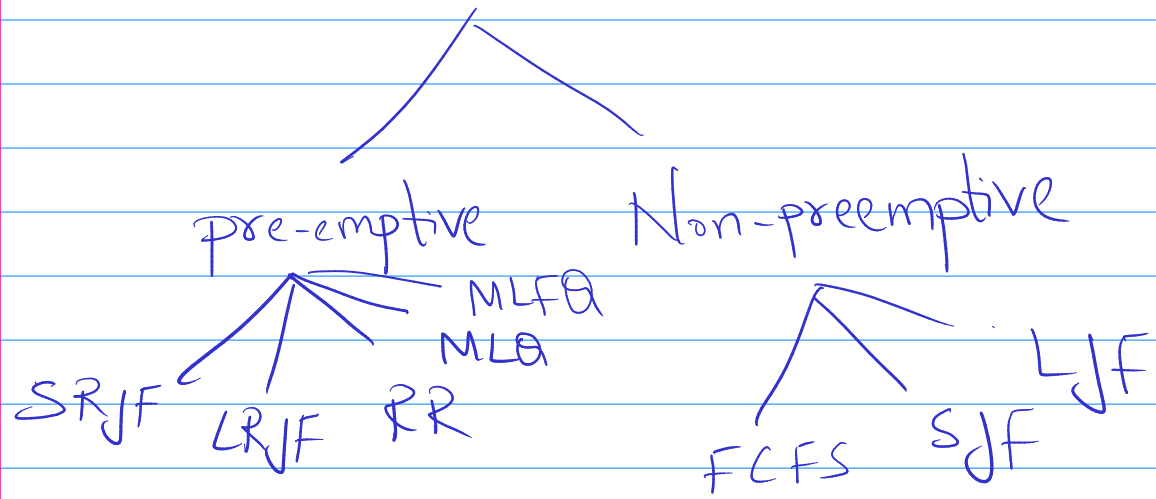
Starvation time

Throughput

fairness

cpu utilization :-

Cpu Scheduling algo's :-



# FCFS :- (FIFO)

↓ Queue

jobn ... job3 job2 job1 → CPU

→

| process name | Arrival Time (sec) | B.T (sec) | R.T |
|---|---|---|---|
| P1 | 0 | 10 | 0 |
| P2 | 1 | 50 | 9 |
| P3 | 2 | 80 | 58 |
| P4 | 4 | 150 | 136 |

Average Response Time = $\frac{0+9+58+136}{4}$

$= 50.75$

Advantage :- Simple to implement

Disadvantage :- R.T is high, throughput is less, cpu utilization

## SJF :-

| Process name | A.T | B.T | R.T |
|---|---|---|---|
| P1 | 0 | 5 ① | 0 |
| P2 | 0 | 10 ② | 5 |
| P3 | 1 | 20 ③ | 14 |
| P4 | 2 | 30 ④ | 33 |

$$A.R.T = \frac{0+5+14+33}{4} = 13$$

→ Disadvantage :-
longest jobs are kept on waiting if the short jobs are entering (or)....

→ Advantage :-
A.R.T is better than FIFO.

LJF :-

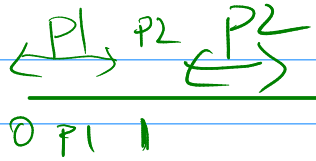
| Process | AT | B.T | R.T |
|---|---|---|---|
| P1 | 0 | 30 | |
| P2 | 1 | 40 | 9 |
| P3 | 1 | 80 | |
| P4 | 2 | 60 | |

Drawback :-

Fairness is not good.

## SRJF :- (Shortest remaining job first) :-

| | A.T | B.T | R.T |
|---|---|---|---|
| P1 | 0 | 10 (1) | 0 |
| P2 | 1 | 5 | 0 |
| P3 | 2 | 15 | 19 |
| P4 | 3 | 6 | 3 |




Drawback :-
Context Switches are more

## → LRJF (Longest remaining job first) :-

| | A.T | B.T. | R.T |
|---|---|---|---|
| P1 | 0 | 30 (29) | 0 |
| P2 | 1 | 40 (39) | 0 |
| P3 | 2 | 60 | 0 |
| P4 | 2 | 20 | 71 |

30-19=11

40 - 19 = 21

60 + 19 = 41

P3 P1

2 3 .....

Drawback :-

- Context Switches are more.

- knowing burst time of a process in advance is not possible in most of the cases so it is not preferable in regular os but in RTOS where Task time is predictable.

- Draw back in SJF, LJF, SRJF, LRJF